

21-1b Shifts in the Budget Constraint

The budget constraint shows the opportunities available to the consumer. It is drawn given the consumer's income and given the prices of the two goods. If the consumer's income or the prices change, the budget constraint shifts. Let's consider three examples of how such a shift might occur.

Suppose first that the consumer's income increases from **\$1,000** to **\$2,000** while prices remain the same. With higher income, the consumer can afford more of both goods. The increase in income, therefore, shifts the budget constraint outward, as in panel (a) of [Figure 2](#). Because the relative price of the two goods has not changed, the slope of the new budget constraint is the same as the slope of the initial budget constraint. That is, an increase in income leads to a parallel shift in the budget constraint.

Figure 2

Shifts in the Consumer's Budget Constraint

In panel (a), an increase in the consumer's income shifts the budget constraint outward. The slope remains the same because the relative price of pizza and Pepsi has not changed. In panel (b), a decrease in the price of Pepsi shifts the budget constraint outward, while in panel (c), a decrease in the price of pizza shifts the budget constraint outward. In these two cases, the slope changes because the relative price of pizza and Pepsi has changed.



Now suppose that the price of Pepsi falls from **\$2** to **\$1** while the consumer's income remains at **\$1,000** and the price of pizza remains at **\$10**. If the consumer spends her entire income on pizza, the price of Pepsi is irrelevant. In this case, she can still buy only **100** pizzas, so the point on the horizontal axis representing **100** pizzas and **0** liters of Pepsi stays the same. But as long as the consumer was buying some Pepsi, the lower price of Pepsi

expands her set of opportunities. The budget constraint shifts outward, as shown in panel (b) of [Figure 2](#). The lower price allows her to buy the same amount of pizza as before and more Pepsi, the same amount of Pepsi as before and more pizza, or more of both goods.

Note that because the slope reflects the relative price of pizza and Pepsi, it changes when the price of Pepsi falls. With a lower price of Pepsi, the consumer can now trade a pizza for 10 liters of Pepsi rather than 5. As a result, the new budget constraint is steeper. The expansion in the consumer's opportunities is represented by a rotational shift rather than a parallel shift.

For our third example, suppose that the price of pizza falls from \$10 to \$5 while the consumer's income remains at \$1,000 and the price of Pepsi remains at \$2. Once again, the lower price expands the consumer's set of buying opportunities and leads to a rotational outward shift in the budget constraint, as shown in panel (c) of [Figure 2](#). Now, with a lower price of pizza, the consumer can now trade a pizza for 2.5 liters of Pepsi rather than 5, and so the budget constraint becomes flatter.

[Figure 2](#) illustrates what happens when a higher income or a lower price expands a consumer's set of opportunities. The opposite occurs when a lower income or a higher price reduces the consumer's opportunities. The pictures look much the same as [Figure 2](#), but with the direction of the arrows reversed. And when more than one change occurs at the same time, we can analyze the overall impact by graphing and comparing the initial and final budget constraints.

QuickQuiz

1. Homer buys pizza for \$10 and Pepsi for \$2. He has income of \$100. His budget constraint will shift inward if
 - a. the price of pizza rises to \$12.
 - b. the price of Pepsi falls to \$1.
 - c. his income rises to \$150.
 - d. the price of pizza, the price of Pepsi, and his income all rise by 50 percent.
2. Marge also buys pizza for \$10 and Pepsi for \$2. She has income of \$200. Her budget constraint will experience a *parallel* outward shift if
 - a. the price of pizza falls to \$5, the price of Pepsi falls to \$1, and her income falls to \$100.

- b. the price of pizza rises to **\$20**, the price of Pepsi rises to **\$4**, and her income remains the same.
- c. the price of pizza falls to **\$8**, the price of Pepsi falls to **\$1**, and her income rises to **\$240**.
- d. the price of pizza rises to **\$20**, the price of Pepsi rises to **\$4**, and her income rises to **\$500**.

Chapter 21: The Theory of Consumer Choice: 21-1b Shifts in the Budget Constraint

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

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